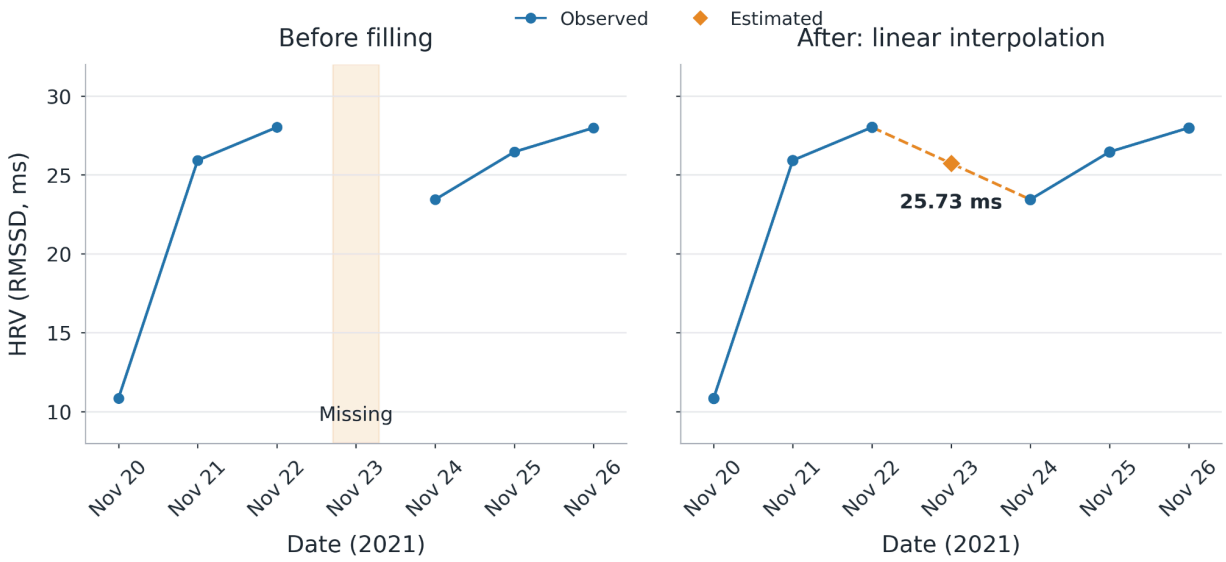
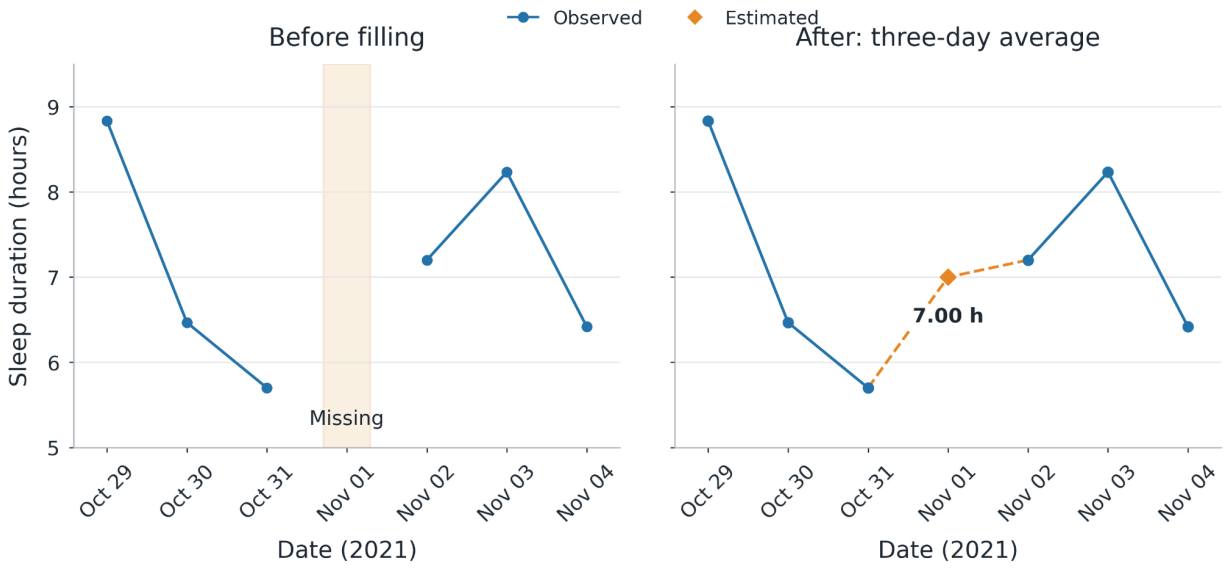
Body Copilot
Aidan Shortley | SE 498 Assignment Part II

Figure 1. HRV before and after filling a missing day



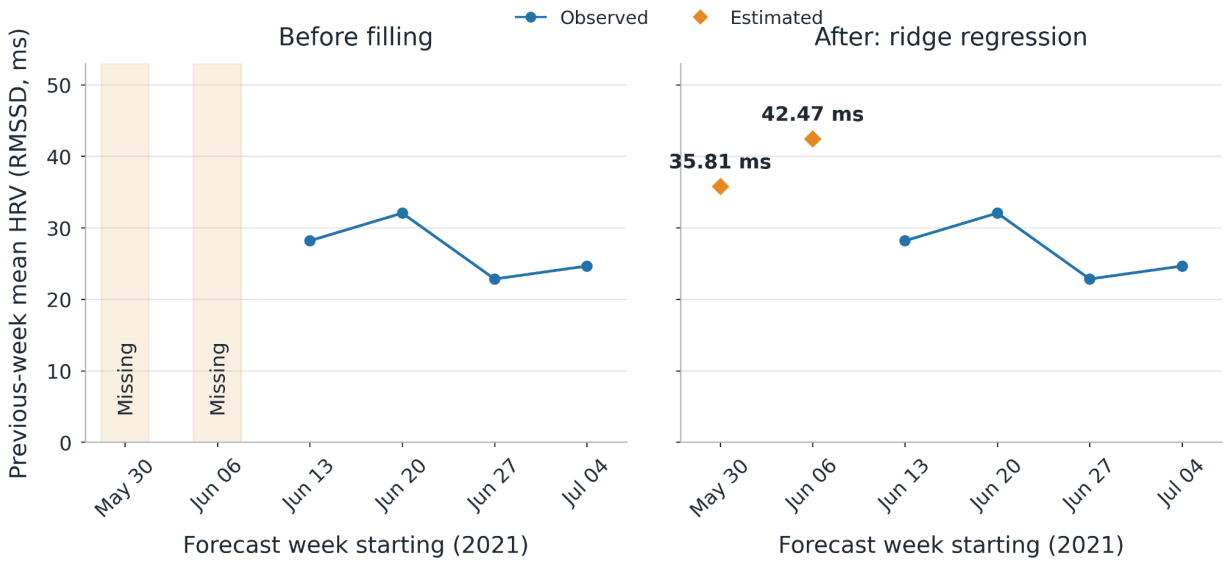
This chart shows one user's daily HRV, measured in milliseconds, with one day missing. This is an example of how we used interpolation to fill in some of the missing values. The orange point is an estimate instead of the observed datapoint.

Figure 2. Sleep duration before and after filling



This chart shows one user's sleep duration, measured in hours. I averaged the previous three days to estimate 7.00 hours of sleep. The orange point is an estimate, not actual sleep duration.

Figure 3. Weekly HRV before and after regression filling



The left chart shows the weekly HRV before filling and the right chart shows ridge regression estimates for HRV for two weeks in a row. Each point represents one weekly feature.

Figures 4-6 compare activity start times with features. Figure 4 is included to explore differences between age groups, but is not included in the current feature extraction. These three plots only include hours containing a recorded start, while the model will predict start versus no start across selected hours.

Figure 4. Age group and recorded start hour

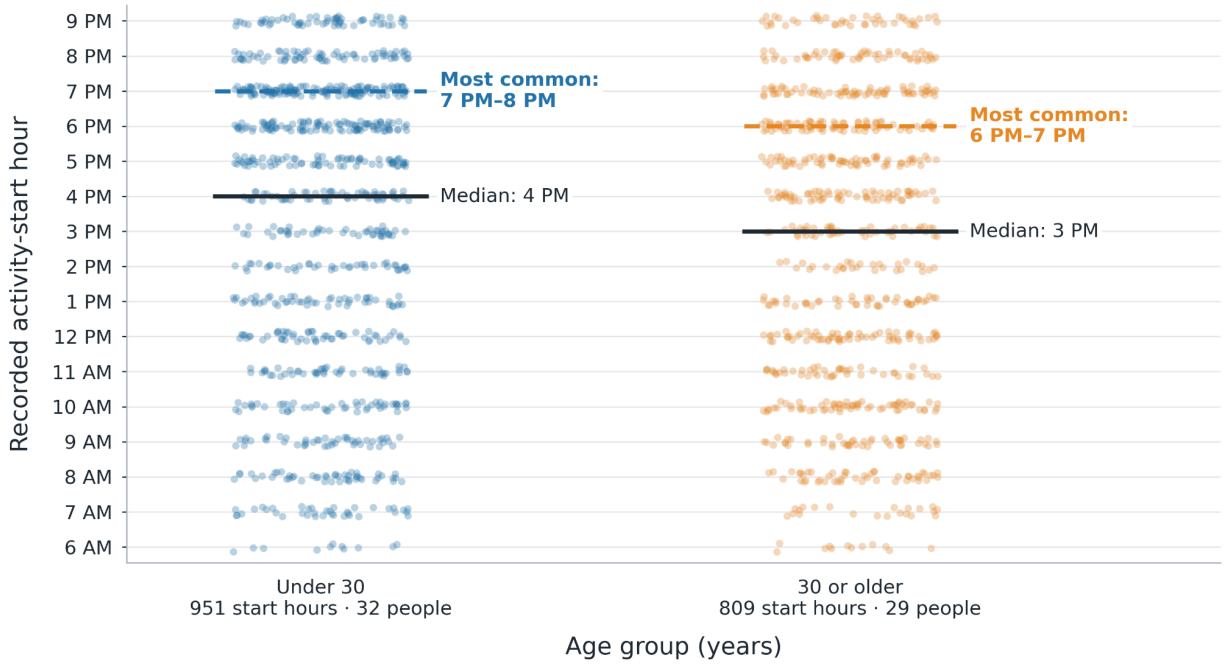


Figure 4 compares activity start hours for participants under 30 and those 30 or older. The chart shows that participants under the age of 30 more commonly had a later recorded activity start time of around 7-8 p.m. while adults over 30 started activity earlier.

Figure 5. Weekday/weekend and recorded start hour

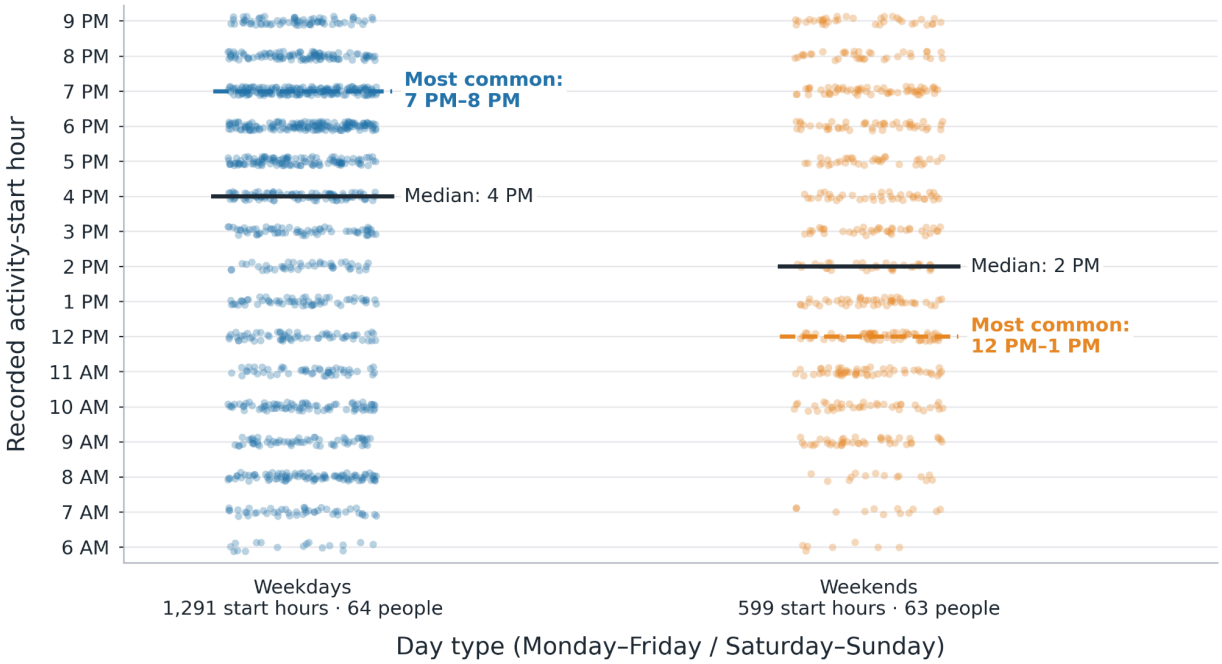


Figure 5 examines when users work out during the week versus the weekend. This can help the model understand patterns during the week and patterns during the weekend. It can be seen that the middle hours are 4 p.m. on weekdays and 2 p.m. on weekends. The most common windows are 7-8 p.m. on weekdays and noon-1 p.m. on weekends.

Figure 6. Previous-week steps and recorded start hour

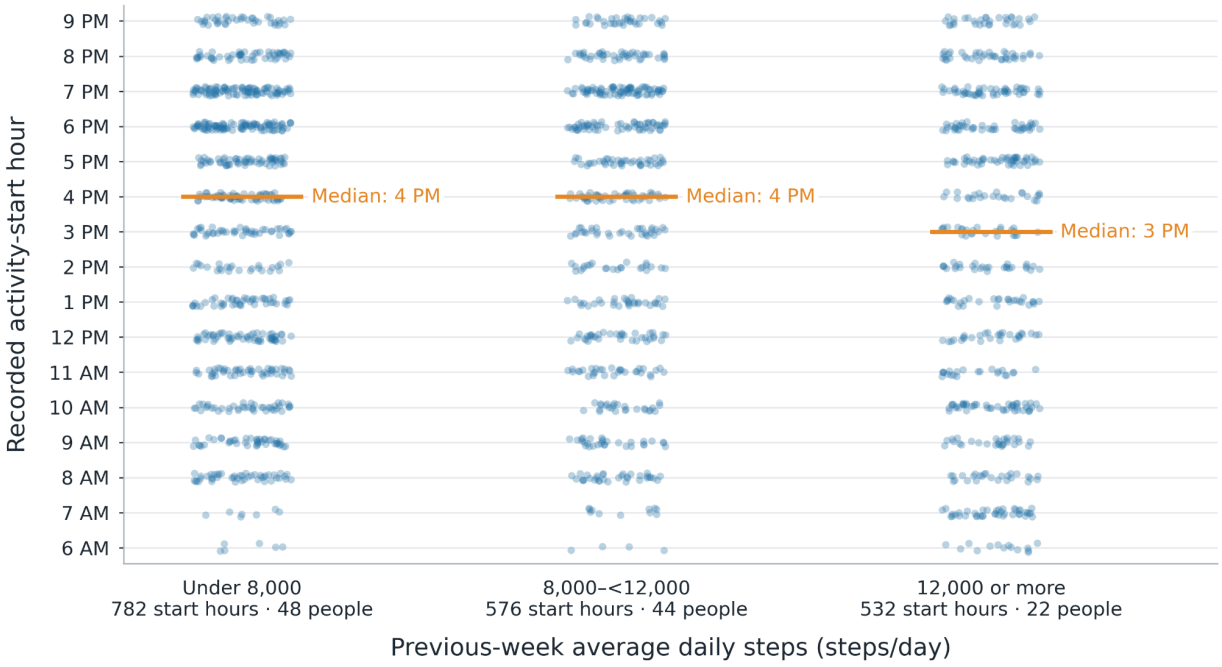
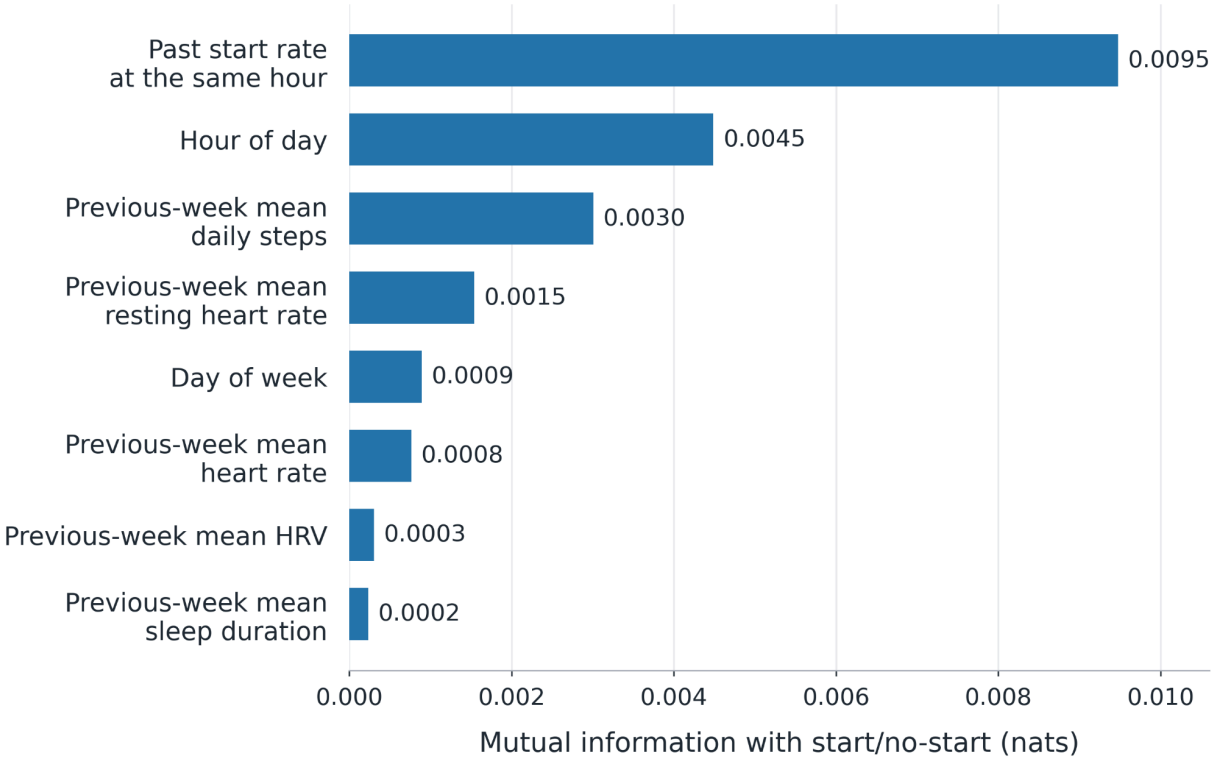


Figure 6. This chart shows how different previous-week average daily step counts relate to when users work out. The trend shows that with 12,000 or more steps, the middle start hour was earlier.

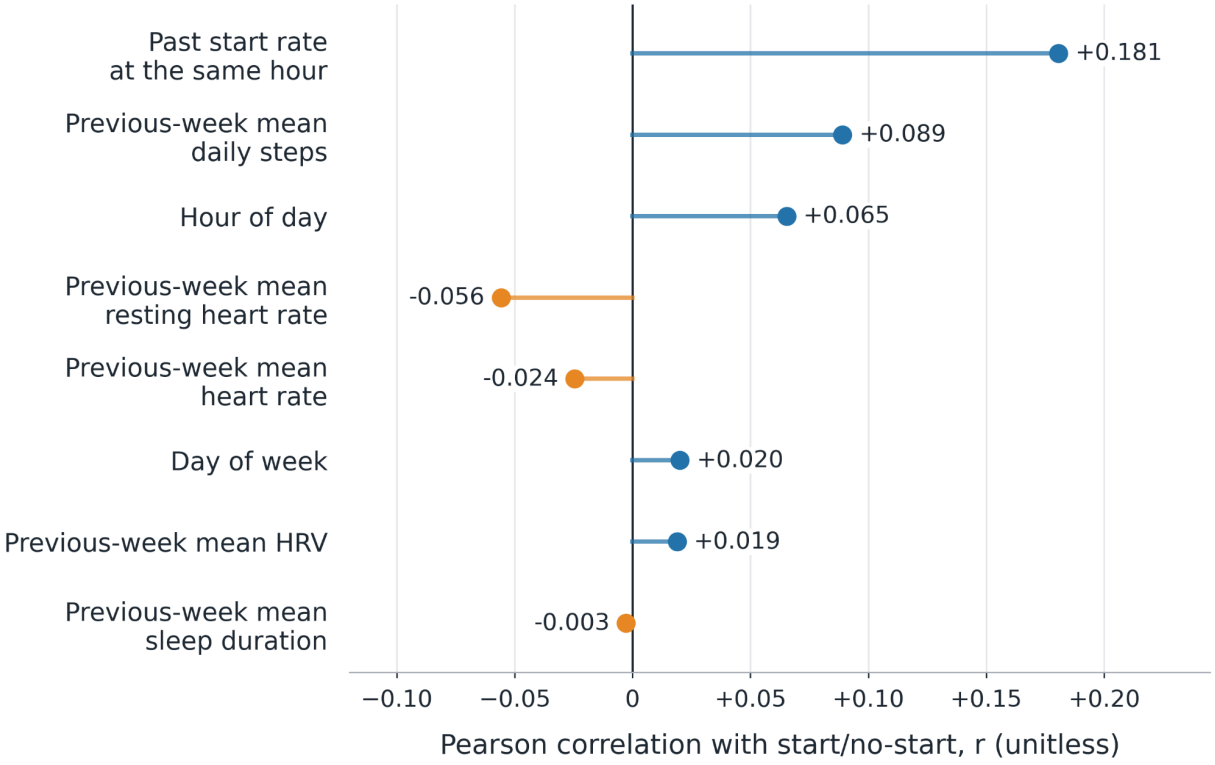
The general trend for age group versus activity start hour shows that younger adults below the age of 30 tend to start activity later while older adults above the age of 30 tend to start activity earlier. Figure 5 shows a trend that people who work out during the week tend to work out later, with the most common window being 7-8 p.m. Participants who worked out on the weekend tend to work out earlier, around 12-1 pm. This could possibly be due to most users having work during the day. The chart shows a substantial increase in workouts during the week between 5-9 pm. Work schedules are a possible explanation. Figure 6 shows how participants with different previous week daily steps relate to recorded activity start. You can see that participants with 12,000 or more previous week average daily steps had a lower middle than participants with less than 12,000 previous week average daily steps. The broad overlap suggests that steps alone do not clearly distinguish workout timing.

Figure 7. Mutual information with activity starts



Past start rate at the same hour had the highest mutual-information score for whether an activity started in that hour, followed by hour of day and previous-week mean daily steps.

Figure 8. Correlation with activity starts



This chart compares each feature with whether a recorded activity started during a candidate hour. 1 = start and 0 = no recorded start. Past start rate at the same hour, previous-week average daily steps, and hour of day had the highest correlation results. This means that a person who had previously started an activity more often during a particular hour was somewhat more likely to start another recorded activity during that same hour. From the wearable measurements, previous-week average HRV had a + 0.019 correlation. In general, all of the correlations are considered weak. These results show associations rather than showing that these features will predict accurately. Later model testing will determine whether the features help.

We plan to use logistic regression to estimate the probability of a recorded start in each candidate hour. This fits because our outcome is binary, either an activity starts (1) or there is no recorded start (0). Logistic regression provides a manageable and interpretable starting model. We will compare models using time and history features with models that also include wearable features in order to test whether adding wearable information improves predictions.